

Semester Project – Guidelines

1. Project Overview

In this project, you are required to identify and implement a computer vision solution to a research question or practical problem of your own choosing. Your project may build upon an existing open-source implementation (for example, one found on GitHub), provided that your team can clearly demonstrate an understanding of the problem, the techniques used, and the reasoning behind every design decision made along the way.

You are expected to draw on the concepts, algorithms, and tools introduced throughout the course's lectures and tutorials, applying them meaningfully to the problem you select. Over the course of the semester, you will report on your progress through three milestone presentations, each corresponding to a distinct stage of your project's development.

Illustrative example:

If a team chose to work on **pedestrian detection for autonomous driving**, the three milestones might unfold as follows:

- **Milestone 1** — introduce candidate datasets (e.g. KITTI, Cityscapes), candidate detection architectures (e.g. YOLO, Faster R-CNN), and any reference GitHub implementations reviewed.
- **Milestone 2** — formally define the specific detection problem the team will address, justify the choice, and outline the implementation plan going forward.
- **Milestone 3** — present the team's results, the challenges encountered, and the approaches explored to improve detection accuracy and robustness.

This example is illustrative only. Teams are free to propose any computer vision problem of interest, provided it is clearly scoped and approved.

2. Presentation Structure

The project will be assessed through three milestone presentations delivered over the course of the semester:

- Each milestone will consist of one PowerPoint presentation.
- Each presentation will be delivered by a single representative of the team (*by default, the team leader*), during the team's assigned presentation slot.
- **The exact starting week of each milestone will be announced on the CMS. Students are responsible for checking the CMS regularly for these announcements.**
- **Milestone 1 presentations will begin the week starting Saturday, September 26, 2026.**
- All team members are expected to understand and be able to discuss the full project, even though only one representative presents; the evaluator may direct questions to any member of the team.
- Every presentation should be clearly organized, delivered professionally, and kept within the allotted time, with slides that are clear, readable, and easy to follow.

3. Milestone Details

Milestone 1 — Problem Exploration & Technical Foundations

Teams should present:

- The general area or motivation behind the computer vision problem being considered.
- Candidate datasets identified for use.
- Candidate models, architectures, and algorithms under consideration.
- Any relevant open-source repositories (e.g. GitHub) reviewed as reference material or starting points.
- The tools and libraries the team intends to use.

Milestone 2 — Problem Definition & Development Plan

Building on Milestone 1, teams should present:

- A clearly defined problem statement: the specific computer vision task the team has chosen to solve.
- Justification for this choice, including how it relates to the datasets, models, and tools explored in Milestone 1.
- A concrete plan describing the team's intended next steps toward implementation.

Milestone 3 — Final Results, Analysis & Improvement Strategies

In the final milestone, teams are expected to present a complete narrative of the project, from motivation through to results. This should include:

- A clear recap of the problem statement, motivation, and objectives.
- The dataset/source of data used, and any preprocessing steps applied.
- The model or algorithm used, explained correctly, with the overall pipeline presented in a clear, logical order.
- Evidence that the implementation is complete, and that the whole team understands the codebase or repository used, not just the presenter.
- A working demonstration of the application, where applicable.
- The evaluation metrics used, and a correct interpretation of the results.
- An honest discussion of the model's limitations.
- Concrete, well-reasoned proposals for future improvement.

Note: this milestone is evaluated on the depth and rigor of your analysis and problem-solving process as much as on your final results — a well-analyzed shortcoming is valued as much as a working improvement.

4. Submission Guidelines

- Presentation slides for each milestone must be submitted on the day of the corresponding presentation.